

Research Paper

Catalytic transformation of plastic waste: Harnessing zeolite for enhanced energy product yield in pyrolysis

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ABSTRACT

Addressing the significant environmental challenge posed by plastic waste demands innovative approaches to its management and conversion into sustainable resources. Catalytic pyrolysis emerges as a promising solution, offering a method to transform plastic waste into high-energy products and contributing to environmental remediation efforts. The study aims to synthesize and characterize the P-zeolite catalyst and assess its influence on the pyrolysis process. X-ray diffraction (XRD), Fourier transform infrared spectroscopy (FTIR), thermogravimetric analysis (TGA), and scanning electron microscopy (SEM) were employed to investigate catalyst structure and properties. Catalytic pyrolysis experiments are conducted using TGA to simulate the process of plastic decomposition and pyrolysis. Additionally, a range of parameters including temperature, catalyst amount, and reaction activation energy have been investigated through both experimental and theoretical analyses. The results demonstrated the pivotal role of the 10 % of synthesized zeolite catalyst in reducing temperature requirements and activation energy by 100 °C and 118 kJ/mol, respectively, compared to thermal pyrolysis conducted without the use of a catalyst.

1. Introduction

In contemporary society, plastics have become integral to the lives of people worldwide. They are utilized in the production of numerous everyday items that are an essential part of our routines. Plastic is used to make a large number of common items which are used on a daily basis. They include food and beverage packaging items, different kinds of packages, toys, furniture, vehicular components, decorations, building materials, and others. Due to the disposable nature of many of these products, they are frequently discarded immediately after use, leading to the accumulation of a substantial portion of municipal solid waste (MSW) elsewhere. Annually, more than 2.01 billion tons of municipal solid waste are generated globally, with approximately 33 % of this waste not being managed in an environmentally safe manner. Among this waste, roughly 70 % finds its way into landfills and other waste

disposal sites, while approximately 19 % undergoes recycling, and about 11 % is utilized for energy recovery [1]. To safeguard the environment, greater emphasis should be placed on recycling and repurposing materials rather than resorting to landfilling or incineration. It's crucial to avoid both landfilling and incinerating plastic waste, as the former poses a risk of soil contamination from degradation by-products, while the latter generates various pollutants harmful to the environment [2]. The rising expenses associated with pollution control measures for incinerators, coupled with the declining favourability of this method in waste management, underscores the necessity for alternative approaches to treating plastic waste [3]. Plastic waste recycling aligns with the four categories of solid waste management. The primary recycling refers to the process of converting plastics back into products resembling their original form. However, secondary recycling also termed mechanical recycling, involves reshaping thermoplastic materials into

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different products suited for less demanding applications. Furthermore, tertiary recycling, specific to plastics, converts waste materials into basic chemicals or fuels. In addition, quaternary recycling involves energy extraction from waste through incineration. Tertiary recycling holds promise in addressing the global glut of plastic waste by mitigating its environmental impact. Tertiary recycling is used to transform polymer materials into low molecular weight chemicals that could be utilized as raw materials in the chemical industry [4]. Among the different technologies, gasification involves converting plastic waste into syngas at temperatures exceeding 700 °C, a process known for its energy intensity despite its capability to handle mixed and contaminated plastics [5]. Hydrothermal liquefaction transforms plastic waste into liquid hydrocarbons under high temperatures (250–374 °C) and pressures, requiring significant energy and precise operational control. Thermal pyrolysis of plastic waste demands substantial energy due to elevated temperatures, resulting in inefficiencies with extended degradation times [6]. Pyrolysis involves subjecting plastic to high temperatures within an inert or vacuum environment, with or without a catalyst. This process leads to the fragmentation of polymer chains, resulting in decomposition, and the generation of gaseous, liquid, and solid new molecules [7]. Lopez et al. [8] showed that pyrolyzing PE waste at 480 °C yielded a total of 65 % gas and liquid products. Moreover, when subjected to a temperature of 550 °C, high-density polyethylene (HDPE) plastic waste yields 17.24 % gas and 80.88 % liquid products [9]. Another investigation focuses on the thermal decomposition of waste plastics and their blends (LDPE, LDPE/PP, and HDPE/LDPE/PP) at 500 °C [10]. Results indicate that pyrolyzing HDPE alone generates approximately 79.8 % liquid, 17 % gas, and 3.2 % solid char, whereas the HDPE/LDPE/PP mixture produces 82.4 % liquid, 13.8 % gas, and 3.8 % char. Pyrolysis of plastic waste involves thermal degradation at temperatures between 500 and 600 °C [11–13]. In some cases, temperatures reach 900 °C [14]. The pyrolysis process depends on temperature [14,15]. The liquid oils produced from thermal pyrolysis often contain impurities and solid residues such as coals. In addition, the thermal pyrolysis of certain polymers is difficult to achieve due to their highly cross-linked structure [15,16]. Catalytic pyrolysis was therefore developed to avoid the problems associated with thermal pyrolysis. Catalysts have found extensive use in pyrolysis operations, not only for upgrading pyrolysis products but also for enhancing hydrocarbon distribution, hindering the production of organic acid by-products, enhancing the thermal degradation process, promoting optimized product distribution and heightened product selectivity. Furthermore, catalysts contribute to generating liquid oil with properties comparable to those of conventional fuels such as diesel and gasoline [17]. Several studies have been carried out on the catalytic pyrolysis of waste plastics using red mud, fly ash, FCC, ZSM-5, HZSM-5, Y zeolites, Fe₂O₃, Al₂O₃, Ca(OH)₂, and natural zeolites [18]. However, the quality and quantity of oils obtained from catalytic pyrolysis depend on the nature of the catalysts [18], acidity, BET surface area, pore size, and crystal structure. Zeolites appear to be the main catalysts used in polymer pyrolysis. Kim et al. [19] conducted an in-depth examination of the HUSY catalyst and compared its performance to that of the HZSM-5 catalyst when used with high-density polyethylene (HDPE) and low-density polyethylene (LDPE) at a temperature of 550 °C. The study focused on evaluating the efficiency and selectivity of the catalysts in converting these common types of plastic waste into valuable hydrocarbon products. Experimental findings indicate that when H-ZSM-5 is employed with LDPE, gas yield reaches 71 % by weight, and with HDPE, it achieves 73 %. Conversely, the utilization of the HUSY catalyst leads to a notable liquid yield, reaching 62 % for LDPE and 41 % for HDPE. Meanwhile, in other studies, only 50 % of the acidity of HZSM-5 after reaction has been recorded after strong deposition of polyaromatic coke [20]. These results underscore the variation in selectivity among zeolite catalysts, attributable to differences in acidity and pore size. Marcilla et al. [19] explored the impact of both microporous and mesoporous catalysts on the pyrolysis of PE and PP waste. They examined three microporous catalysts (HUN-ZSM-5, C-ZSM-5, and zeolite β) alongside

three mesoporous catalysts (Al-MCM-41, KFS-16B, and Al-SBA-15 (wo)). Their findings revealed that HUN-ZSM-5 yielded the highest gasoline fraction at 96 %, with β -zeolite following closely behind at 85 %. Saeung et al. [21] investigated the effects of Zeolite and MgO on the catalytic degradation of HDPE, LDPE, PP, and biodegradable plastic polylactide (PLA). The results showed no significant impact on the temperature range. However, the conversion rate was notably high with these catalysts, with MgO exhibiting significant coke deposition, likely due to differences in catalyst properties. Conversely, Cocchi et al. [22] observed a small temperature shift from 472 to 425 °C using HZSM-5 during the catalytic pyrolysis of marine plastic litter in a bench-scale reactor. In conclusion, the unique characteristics of zeolites, including their small crystal size and increased external surface area, enable the efficient reaction of large molecules at external sites while enhancing the internal diffusion rate of smaller molecules within micropores. These attributes afford zeolites numerous advantages over other catalytic materials for polyolefin cracking, such as heightened activity even at low temperatures and improved selectivity. In addition to their attractive properties for waste plastic degradation, zeolites are cost-effective compared to other materials as they can be synthesized from cost-free resources. Economically, a study by Chang et al. [23] on the feasibility of catalytic pyrolysis of plastics using zeolites estimated that the cost of zeolite catalysts could account for a significant portion of the overall process cost, especially in large-scale applications. Moreover, Shemfe et al. [24] reported that the regeneration and recycling of spent zeolite catalysts can increase operational costs, but these costs may be offset by the reduction in energy consumption due to lower reaction temperatures. Consequently, using zeolite synthesized from cost-free resources as well as reducing the required consumption during the pyrolysis process represents a promising opportunity to reduce the overall cost of the process.

Thermogravimetric analysis (TGA) serves as a common method for characterizing mass loss during pyrolysis. By conducting TGA experiments with small sample sizes and variable heating rates, researchers can reliably explore the kinetic regime of conversion mechanisms. Moreover, TGA enables the determination of activation energy, which signifies the minimum energy required to initiate the thermal decomposition of the polymeric material. This pivotal parameter plays a crucial role in the analysis of plastic pyrolysis, utilizing curves obtained from TGA. The calculation of non-isothermal kinetic parameters encompasses two main approaches: fitted model methods and iso-conversive methods. Each of these methods possesses specific limitations and is employed in a complementary manner [25]. Iso-conversive methods offer a means to evaluate the evolution of the reaction's effective kinetic energy without necessitating the assumption of a reaction model. This approach reflects a dependence on the degree of conversion and indirectly captures the complexity of the reactions and mechanisms inherent in plastic pyrolysis processes. Several iso-conversive kinetic methods are available, including the Friedman differential method, the Ozawa-Flynn-Wall (FWO) integral method, the Kissinger-Akahira-Sunose (KAS) integral method, and the Vyazovkin non-linear integral method [26,27]. Vimalathithan et al. investigated the catalytic effects of various clay nanoparticles on the pyrolysis of polypropylene (PP). They introduced four types of nanocomposites into PP, each comprising one of four distinct clay nanoparticles (Cloisite Na⁺, Nanofil5 Nanoclay Mica, Bentone 38, and Nanofil5). By applying the Ozawa method to calculate the activation energy (E_a) at different conversion levels, they observed a notable reduction in the E_a required to initiate decomposition, particularly evident in the case of PP/Cloisite Na⁺, where it decreased from approximately 158 to 124 kJ/mol [28]. In a similar investigation, researchers conducted a kinetic analysis at varying heating rates, omitting the use of specific models such as Ozawa-Flynn-Wall, Kissinger-Akahira-Sunose, and Friedman. This analysis aimed to evaluate the thermal decomposition behavior and activation energy (E_a) of polypropylene (PP), low-density polyethylene (LDPE), and polyvinyl chloride (PVC). The researchers ranked the complexity of

pyrolysis processes in the following order: PVC < PP < LDPE. For instance, E_a values for PP obtained from all three models ranged from 103.02 to 110.75 kJ/mol [29]. Additionally, the study by Chi et al. [30] demonstrated a significant decrease in activation energy (E_a) and maximum temperature (T_{max}) when MCM-41 and Al-MCM-41 catalysts were present. The study aims to synthesize and characterize P-zeolite, followed by an investigation into its catalytic activity on high-density polyethylene (HDPE). Through catalytic pyrolysis experiments conducted under different conditions, the research seeks to evaluate the performance of P-zeolite and establish relationships between its catalytic effectiveness, intrinsic properties, and the specific attributes of the polymer.

2. Materials and methods

2.1. Materials

Kaolinite clay with the chemical composition of Table.1 purchased from Sibelco Company, UK, was thermally treated at 750 °C for 5 h using a heating rate of 5 °C/min to produce a metakaolin with high activity, which was used for the synthesis of zeolite. NaOH (purity $\geq 99\%$, pellets, anhydrous), HCl (98 %), silicate de sodium Na_2SiO_3 (89 %), KOH (99 %).

2.2. Zeolite synthesis

The P-zeolite used in this study was synthesized from metakaolin by a two-stage hydrothermal reaction, as shown in Fig. 1. In the first hydrothermal reaction, 3 g of metakaolin are mixed with a 16 ml solution of 8 M NaOH and heated to 200 °C for 3 h in a stainless-steel autoclave. Following filtration, the pH was adjusted to 7 with deionized water, and the mixture was dried for 24 h at 90 °C. The resulting mixture was mixed with 60 ml of 0.2 M hydrochloric acid solution, stirred for 30 min at room temperature, then the synthesized solids were separated by filtration. The pH was then adjusted to 7 with NaOH, and the filtrate produced an amorphous white aluminosilicate gel. In the second hydrothermal reaction, the aluminosilicate gel was mixed with 45 ml of 1 M NaOH solution while stirring at room temperature to homogenize it. The mixture was then heated in a stainless-steel autoclave at 90 °C for 24 h to produce P-zeolite. Finally, the resulting products were washed with deionized water and dried at 90 °C for 24 h.

2.3. Catalyst characterization

X-ray diffraction (XRD) analyses of the prepared catalysts were conducted on a Bruker diffractometer D8 Advance utilizing Cu-K α radiation. These analyses were performed at room temperature, covering diffraction angles ranging from 10 to 70°. Fourier-transform infrared spectroscopy (FT-IR) measurements were carried out using a Frontier Perkin spectrometer, with spectra recorded across wavelengths ranging from 4000 to 400 cm^{-1} . The surface morphologies of the prepared catalysts were observed using a ZEISS Sigma 300 VP scanning electron microscope (SEM). Thermal analyses were conducted for both catalysts and plastics utilizing a Netzsch STA 449 Jupiter F3 coupled TG-DSC analyzer. The catalysts' surface area and pore size were determined using Brunauer–Emmett–Teller (BET) analysis with a Micromeritics Gemini VII 1014 instrument. The particle size distribution of the synthesized zeolite was measured using the laser diffraction method. Initially, the zeolite powder underwent 1-minute ultrasonication at 300

W. The refractive index used was 1.52 with an absorption index of 0.1 for the zeolite, while for water, the refractive index of 1.33 was employed.

2.4. Catalytic pyrolysis tests

The high-density polyethylene (HDPE) plastic was procured from Casablanca. The choice of these specific plastic materials was driven by their predominant contribution to plastic waste generation in Morocco. Subsequently, thermogravimetric analysis (TGA) was conducted on blended HDPE plastic, combined with varying proportions of catalysts. The catalyst-plastic blends underwent additional processing, involving pulverization into powder (<100 nm) using a ball-milling machine (Retsch MM 480) for 3 h at a frequency of 20 Hz/sec. Following this, the preparations underwent TGA analysis to simulate the pyrolysis process conditions.

2.5. Pyrolysis kinetics modeling

The pyrolysis of plastics involves the intricate breakdown of long-chain molecules, yielding light hydrocarbons or radicals. While the reactions' full complexity defies complete modeling, simplifying the process through a one-step reaction has proven effective across various plastic types [31,32]. To calculate the activation energy (E_a) of the reaction, it is necessary to evaluate the E_a for each conversion (x). The average of the E_a values calculated for each conversion gives the overall E_a of the reaction. The variable x can be defined by equation (1).

$$x = \frac{m_0 - m_t}{m_0 - m_\infty} \quad (1)$$

The terms m_0 , m_t and m_∞ represent the initial mass, the sample mass at time t and the final mass respectively. The conversion rate is expressed as shown in equation (2) [33].

$$\frac{dx}{dt} = K(T) \times f(x) \quad (2)$$

Furthermore, $K(T)$ is a temperature-dependent function defined by the Arrhenius equation (3) [34], in which E_a (J/mol) represents the activation energy, A is the pre-exponential component known as the frequency factor, and R is the universal gas constant (8.314 J/mol.K).

$$K(T) = A \times e^{-\frac{E_a}{RT}} \quad (3)$$

In addition, $f(x)$ is selected from the most frequently encountered reaction mechanisms, as shown in Table 2 [35], where $g(x)$ represents the integrated form of the reciprocal of the reaction model $f(x)$, with respect to conversion dx , according to equation (4).

$$g(x) = \int_0^x \frac{dx}{f(x)} \quad (4)$$

The aim is to identify the reaction model $f(x)$ using Criado's method, defined by equation (5). This method involves a graphical curve fitting comparing all theoretical reaction models $f(x)$, presented in Table 2 and normalized by equation (6), with the experimental function $e_{\text{experimental}}$ defined by equation (7).

$$\frac{z(x)}{z(0.5)} = \frac{f(x)g(x)}{f(0.5)g(0.5)} = \frac{(T_x^2)}{(T_{0.5}^2)} \frac{(\frac{dx}{dt})_x}{(\frac{dx}{dt})_{0.5}} \quad (5)$$

Table 1

Chemical composition of Kaolin clay and synthesized zeolite.

Elements	SiO ₂	Al ₂ O ₃	CaO	Fe ₂ O ₃	K ₂ O	MgO	Na ₂ O	P ₂ O ₅	SiO ₂ /Al ₂ O ₃
Kaolin	51.88	38.79	0.07	0.67	2.58	0.29	0.02	0.05	1.33
Synthesized zeolite	60.78	29.22	0.16	0.49	1.32	0.78	5.81	***	2.08

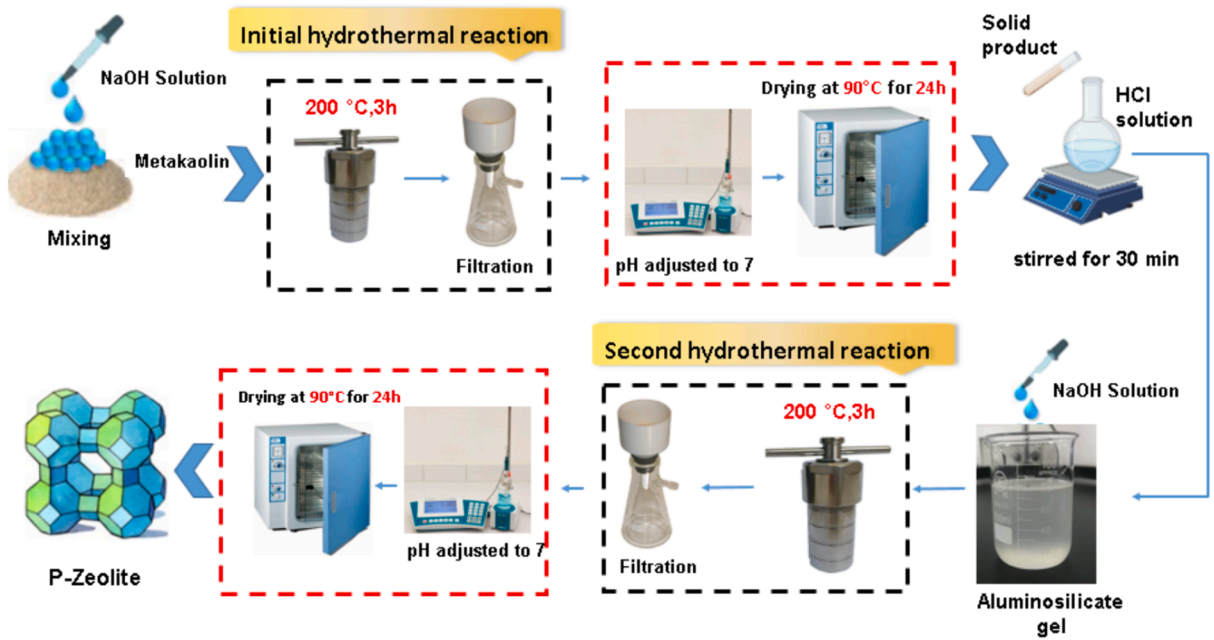


Fig. 1. Illustration of the P-Zeolite synthesis process.

Table 2

Reaction model for the most common reaction mechanism.

Mechanism	Symbols	f(x)	g(x)
First-order	F1	$(1-x)$	$-\ln(1-x)$
Second-order	F2	$(1-x)^2$	$(1-x)^{-1}-1$
Third-order	F3	$(1-x)^3$	$(1-x)^{-1}-1$
			2
Power law	P2	$2x^{1/2}$	$x^{1/2}$
Power law	P3	$2x^{2/3}$	$x^{1/3}$
Power law	P4	$4x^{3/4}$	$x^{1/4}$
One-dimensional diffusion	D1	$\frac{1}{2}(x)$	x^2
Two-dimensional diffusion	D2	$(-\ln(1-x))^{-1}$	$((1-x)\ln(1-x)) + x$
Three-dimensional diffusion	D3	$\frac{3(1-x)^{2/3}}{(2(1-(1-x)^{1/3}))}$	$(1-(1-x)^{1/3})^2$
Ginstling-Brounshtein	D4	$\frac{3}{2}((1-x)^{-1/3}-1)$	$1-\frac{2x}{3}-(1-x)^{2/3}$
Contracting cylinder	R2	$2(1-x)^{1/2}$	$(1-(1-x)^{1/2})$
Contracting sphere	R3	$3(1-x)^{2/3}$	$(1-(1-x)^{1/3})$
Avrami-Erofe'ev	A2	$2(1-x)(-\ln(1-x))^{1/2}$	$(-\ln(1-x))^{1/2}$
Avrami-Erofe'ev	A3	$3(1-x)(-\ln(1-x))^{2/3}$	$(-\ln(1-x))^{1/3}$
Avrami-Erofe'ev	A4	$4(1-x)(-\ln(1-x))^{3/4}$	$(-\ln(1-x))^{1/4}$

$$f_{theoretical} = \frac{f(x)g(x)}{f(0.5)g(0.5)} \quad (6)$$

$$f_{experimental} = \frac{(T_x^2)}{(T_{0.5}^2)} \frac{(\frac{dx}{dt})_x}{(\frac{dx}{dt})_{0.5}} \quad (7)$$

$T_{0.5}$ and $(\frac{dx}{dt})_{0.5}$ correspond respectively to the temperature and conversion rate when x reaches 0.5. Division by 50 % is used to normalize the value of $z(x)$. The coefficient of determination (R^2) is then calculated by comparing the experimental curves with the theoretical ones. The models with the highest regression coefficients (R^2) are finally selected as the most appropriate reaction models. Determination of the

activation energy E_a (J/mol), requiring the introduction of the concept of isoconversion. The kinetic equation, Eq. (2), is modified by applying the logarithmic function on both sides and differentiating with respect to $1/T$ at constant conversion x , leading to equation (8) [36].

$$\left(\frac{\partial \ln(dx/dt)}{\partial T^{-1}}\right)_x = -\frac{E}{R} \quad (8)$$

The Kissinger-Akahira-Sunose (KAS) method has rectified certain deviations from simpler methods such as Ozawa-Flynn-Wall, using a Coats-Redfern approximation [37]. For the KAS method, the Arrhenius equation remains the following expression:

$$\ln\left(\frac{\beta}{T^2}\right) = \ln\left(\frac{AR}{E_a g(x)}\right) - \frac{E_a}{RT} \quad (9)$$

Plotting $\frac{\beta}{T^2}$ versus $\frac{1}{T}$ for a constant conversion produces a linear curve. The slope of this curve provides the value of E_a .

3. Results and discussion

3.1. Characterization of the synthesized zeolite

The X-ray diffraction analysis of the clay mineralogical phase (Fig. 2a) indicates the presence of kaolinite [$\text{Si}_2\text{O}_5\text{Al}_2(\text{OH})$], with distinct peaks detected at various angles according to the ASTM card 14-164. Quartz (SiO_2), illite, and anatase are also present, albeit in smaller quantities. The prevalence of kaolinite is apparent in the diffraction pattern, suggesting its higher concentration relative to other minerals. After calcination at 750 °C, X-ray diffraction patterns (Fig. 2b) reveal significant changes in the crystalline structure of the sample. Subsequent to this transformation, the X-ray spectrum indicates the disappearance of diffraction peaks associated with specific crystalline planes of kaolinite, typically occurring around 12.15° and 25.03°. This observation confirms the alteration or transformation of the crystalline structure of kaolinite. Furthermore, an enlargement of the amorphous band is observed, supporting the conversion of kaolin into metakaolin, which is generally characterized by a disordered and amorphous structure. (Fig. 2c) depicts the X-ray diffraction pattern of the synthesized zeolite. Prominent peaks are observed at 7.18°, 10.14°, 23.94°, 27.05°, and 29.88°, corresponding to the most common peaks for P-zeolite and

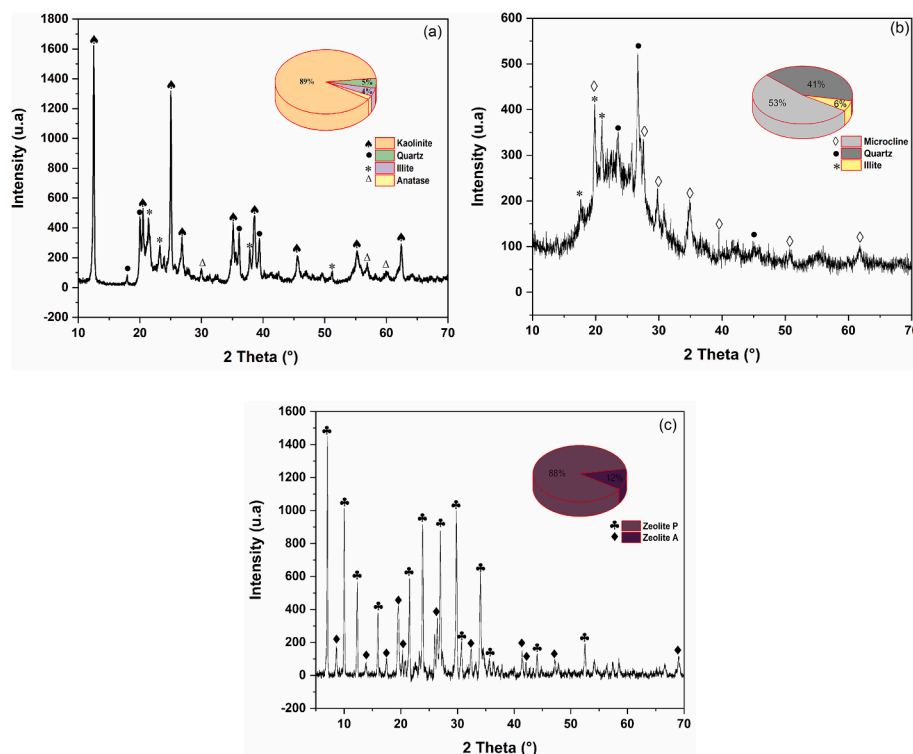


Fig. 2. X-ray diffraction pattern of kaolin (a), metakaolin (b) and synthesized Zeolite (c).

aligning with the (100), (110), (311), (321), and (410) crystallographic planes, respectively. Additionally, peaks associated with the presence of A-zeolite are also identified. Quantitative analysis of the XRD pattern confirms that P-zeolite is the predominant component, constituting 88 % of the composition.

The FT-IR spectra of kaolin, metakaolin, and synthesised Zeolite are shown in Fig. 3. The spectrum of kaolin (Fig. 3a) shows broad bands at $3600\text{--}3740\text{ cm}^{-1}$ corresponding to the stretching mode of the OH. The bands at 1100 and 902 cm^{-1} are linked to the stretching vibrations of the Al-O octahedral sheets and Si-O bonds respectively. Similarly, characteristic bands at 792 and 750 cm^{-1} corresponding to the tetrahedral coordination of aluminum were detected [38]. In addition, two bands at 542 and 696 cm^{-1} attributed respectively to the bending of Si-O-Al and Si-O-Si [38]. Fig. 3b shows the FT-IR spectrum of metakaolin obtained after heat treatment of kaolin. The absorption bands at 1062 and 794 cm^{-1}

cm^{-1} are associated with the asymmetric stretching vibrations of the Si-O and Al-O bands, respectively. The disappearance of the characteristic OH bands after calcination of the kaolin confirms its transformation into metakaolin, which is in line with the results obtained by XRD. Fig. 3c shows the FT-IR spectrum of prepared Zeolite. The peak at 552 cm^{-1} is attributed to the double ring inside the tetrahedral structure [39]. The most significant vibration in the 1000 cm^{-1} region is attributed to T-O stretching (where T can be an Al or Si ion) which involves motions that are predominantly associated with oxygen atoms or described as an asymmetric O-T-O stretching mode [40].

The morphology of the Kaolin, metakaolin and synthesized zeolite was observed using a scanning electron microscope, the results are shown in Fig. 4. SEM image of kaolin shows crystals in the form of pseudo-hexagonal platelets [41], whereas metakaolin has an indeterminate shape [42]. Furthermore, the morphology of the synthesized zeolite (Fig. 4c, d) shows the presence of crystalline spheres and other particles form related to the co-existence of A-zeolite. The clearly defined morphology proves the presence of P-zeolite, confirming the XRD and FT-IR results. Moreover, the results obtained are in line with prior studies [39], which also highlighted challenges in synthesizing pure P-zeolite.

The thermogravimetric analysis (TGA) curves (Fig. 5) of the kaolin exhibit two distinct weight loss regions attributed to the removal of water molecules. Below $200\text{ }^{\circ}\text{C}$, there is a loss of physisorbed water, while above $450\text{ }^{\circ}\text{C}$, dehydroxylation of coordinated and structural water occurs. Typically, clay materials contain three types of water molecules within their structure. Physisorbed and interlayer water are loosely bound and mobile, thus removable through heat treatment below $200\text{ }^{\circ}\text{C}$. Water molecules situated in the first coordination sphere of the interlayer ions are strongly bonded and necessitate higher temperatures, typically in the range of $300\text{--}500\text{ }^{\circ}\text{C}$, for their removal. Lastly, structural hydroxyl groups can condense and dehydrate within the temperature range of $500\text{--}800\text{ }^{\circ}\text{C}$. In this study, the weight loss at low temperatures can be attributed to physisorbed water, while the weight loss at high temperatures is primarily due to the dehydration and dehydroxylation of the clay sheet [43]. The thermogravimetric curve of

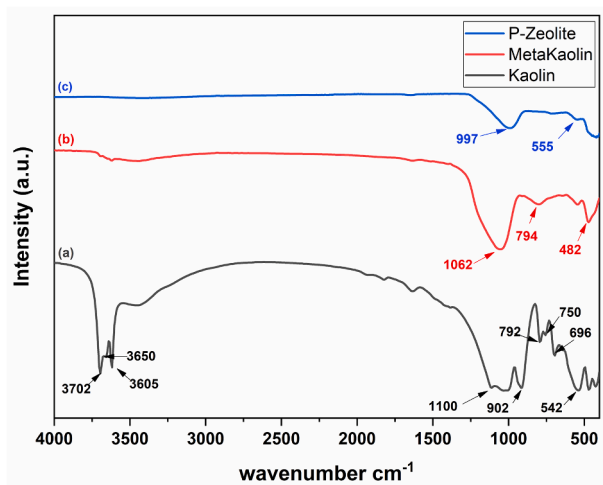


Fig. 3. FT-IR spectra of kaolin (a), metakaolin (b), and synthesized Zeolite (c).

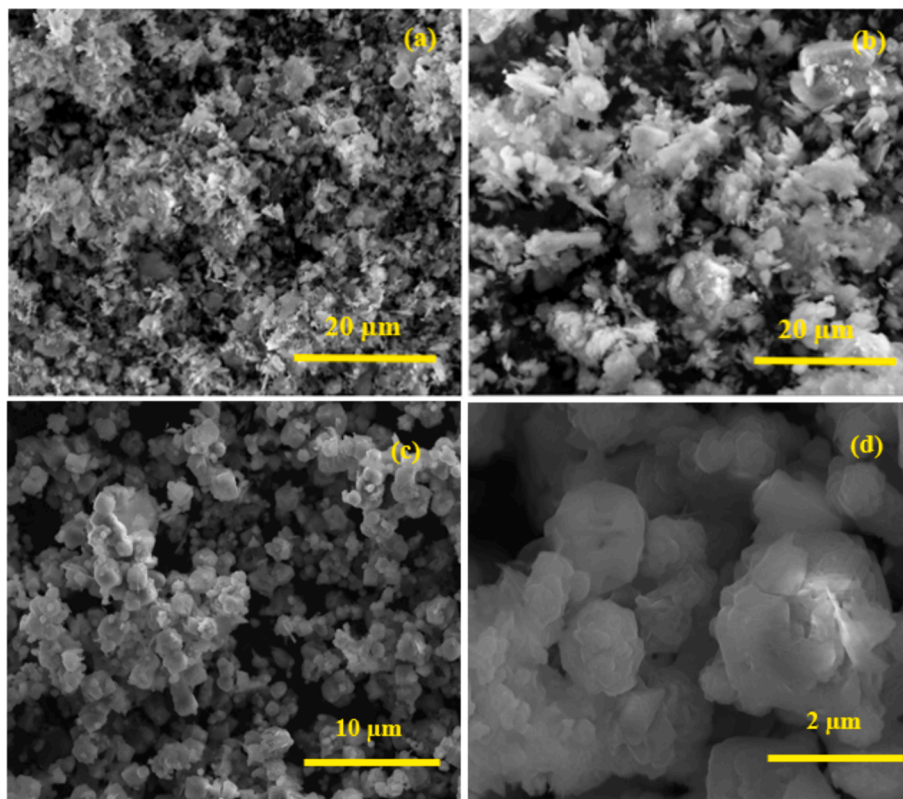


Fig. 4. SEM images of kaolin (a), metakaolin (b) and synthesized zeolite (c,d).

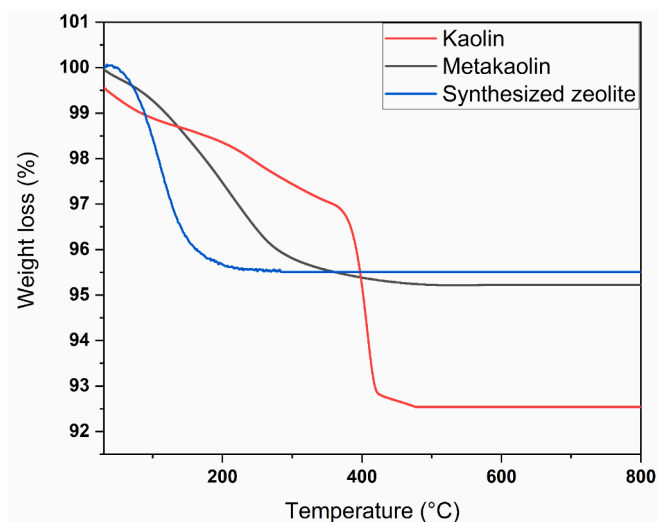


Fig. 5. TGA curves of Kaolin, Metakaolin, and synthesized Zeolite.

metakaolin exhibits a minor weight loss of approximately 5 %, attributed to the elimination of physisorbed water and other volatile components within the sample. This weight reduction occurs at temperatures below 300 °C.

The synthesized zeolite exhibits a gradual weight loss in the temperature range of 30 °C to 200 °C, primarily attributed to the removal of physically adsorbed water and subsequent dehydration of the α and β cages [43]. Following this, no significant additional weight loss is observed. These findings indicate that the synthesized zeolite is a promising candidate for high-temperature applications.

The synthesized zeolites display Type IV isotherms according to the IUPAC classification (Fig. 6a), indicating their nature as microporous

materials with high surface areas. This type of isotherm demonstrates rapid gas uptake at low relative pressures, underscoring efficient micropores filling. The presence of a Hysteresis Loop further indicates the coexistence of mesoporosity alongside micropores. The BJH analysis (Fig. 6b), confirms a distribution of mesoporous and potentially some micropores, broadening the material's pore size distribution. Additionally, Fig. 6c illustrates a granulometric distribution centered around a 3 μm diameter, highlighting uniform particle sizes. This uniformity is advantageous for applications requiring consistent porosity and optimized adsorption properties. As showing in Table 3, the synthesized zeolite exhibits impressive characteristics, including a BET surface area of 134.32 m^2/g , a cumulative pores volume of 0.81 cm^3/g , and an average pore diameter of 3.66 nm. These characteristics highlight Zeolite P as a highly porous material with significant surface area and porosity. These attributes render it well-suited for diverse applications including catalysis, adsorption, and separation processes.

3.2. Characterization of plastic feedstock

HDPE was selected as a primary candidate for investigating the catalytic influence of zeolite on pyrolysis because of its widespread use, uncomplicated chemical structure, and thermal resilience. Exploring zeolite's impact on HDPE pyrolysis yields valuable insights into plastic degradation mechanisms and kinetics, which often extend to other types of plastics. Thus, employing HDPE as a model system provides a robust foundation for understanding the broader implications of zeolite catalysis in efforts to recycle and valorize plastic waste. However, before starting the catalytic test, the plastic feedstock was analysed using FTIR, XRD, and TGA analysis. Fig. 7a illustrates the FT-IR spectra of pure HDPE polymer, revealing distinct double bands at 2912 cm^{-1} and 2847 cm^{-1} , corresponding to the stretching vibrations of $-\text{CH}_2$ and $-\text{CH}$, respectively [44]. Additionally, bands observed at 723 cm^{-1} and 1458 cm^{-1} were attributed to the rocking and bending vibrations of CH_2 , respectively [45].

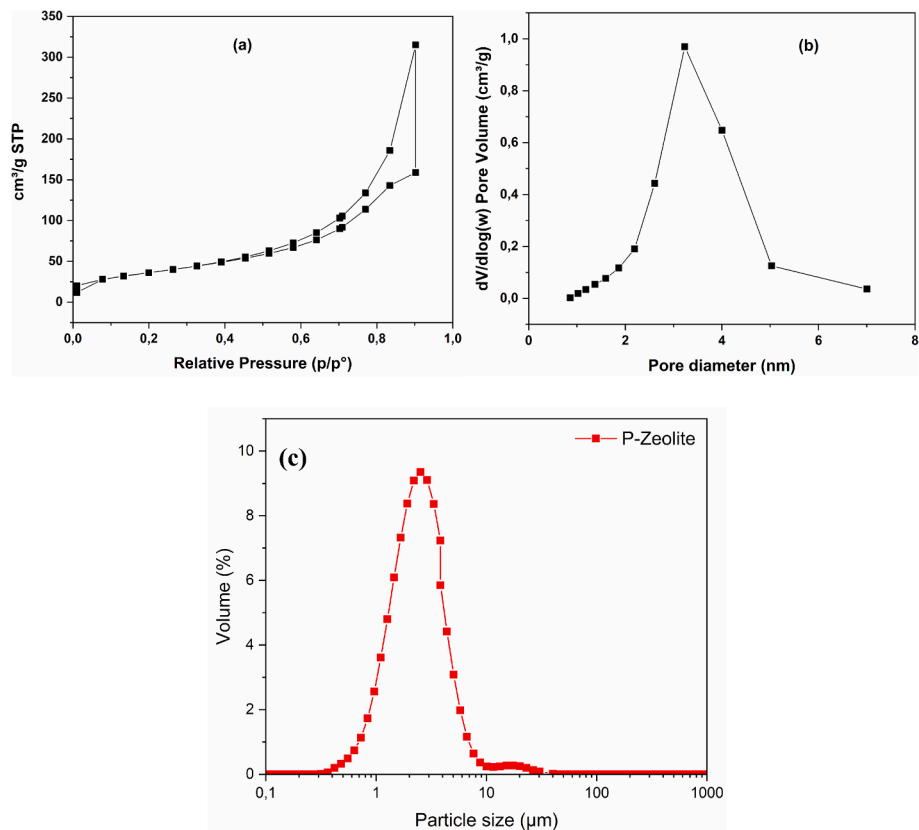


Fig. 6. Textural properties of synthesized P-Zeolite (a) BET plot surface area (b) BJH pore and diameter size, and (c) particle size distribution.

Table 3
Textural properties of synthesized zeolite.

BET surface area (m ² /g)	Cumulative pores volume (cm ³ /g)	Average pore diameter(nm)
134.32	0.81	3.66

The TGA curve of HDPE (Fig. 7b) illustrates distinct weight loss regions corresponding to different thermal decomposition processes. Initially, there's a gradual weight loss attributed to the removal of physically adsorbed water at lower temperatures. Subsequently, the degradation of virgin HDPE begins at 424 °C and concludes at 514 °C. At approximately 488 °C, 50 % weight loss (T₅₀ %) is observed. This weight loss is a result of the degradation of the polymer chains, involving the breaking of carbon–carbon bonds in the polymer backbone. This process releases volatile organic compounds and forms char residue. The observed pattern in HDPE decomposition by TGA aligns with similar reported trends [46,47]. The TGA curve offers valuable insights into the thermal stability and decomposition behavior of HDPE, crucial for various industrial applications, including fuel production via pyrolysis processes.

The X-ray Diffraction (XRD) spectra of neat HDPE (Fig. 7c) show a prominent peak at 21.5°, along with two moderately strong peaks at 23.8° and 36.5°, corresponding to interplanar spacing of 3.841 Å, 3.537 Å, and 2.383 Å, respectively. These peaks are associated with the (1 1 0), (200), and (020) lattice planes of the orthorhombic structure. Additionally, the intensity of the peaks corresponding to the (1 1 0) and (200) planes was analyzed to determine the crystalline thickness and crystallinity of the HDPE. The calculated crystalline size is 28.19 nm, and the degree of crystallinity is 49.67 %. These results align with other reported findings for HDPE [48,49].

3.3. The catalytic effect of zeolites on the degradation temperature of polymers

Thermogravimetric Analysis (TGA) was employed to explore the effect of synthesized zeolite on HDPE onset temperature, peak degradation temperature, and weight loss. These parameters provided insights into catalytic mechanisms, optimal conditions, and potential enhancements for plastic pyrolysis efficiency in waste management and resource recovery. For that, plastic samples mixed with various amounts of synthesized zeolite catalyst underwent controlled temperature increases from 25 °C to 700 °C at a constant rate. Comparing TGA curves of HDPE with and without synthesized zeolite catalyst (Fig. 8) revealed changes in reaction pathways, transitioning from single-step degradation to multiple stages of weight loss. This could be related to the interesting properties of the studied catalyst, such as high surface area, well-defined pore structure, and acidic sites, which facilitate efficient polymer adsorption, diffusion, and catalytic cracking. These attributes allow the zeolite to provide an alternative reaction pathway with lower energy barriers and stabilize transition states and intermediates, resulting in lower activation energy and degradation temperature. Consequently, the presence of zeolite enhances the decomposition process by promoting cracking reactions, that may promote the formation of smaller hydrocarbon fragments such as pentane, 2-methyl-1-pentene, 2,4-dimethyl-1-heptene, and propane[50]. This catalytic activity is facilitated by the adsorption of HDPE molecules onto the catalyst surface, followed by the activation of carbon–carbon bonds and the subsequent fragmentation of the polymer chains.

Table.4 presents the degradation temperatures of HDPE across three degradation stages in the presence of various percentages of synthesized zeolite catalyst. Analysis of the results indicates that the addition of catalyst reduces the degradation temperature significantly, a trend observed up to the addition of 10 % catalyst. However, beyond this percentage, this trend is not respected, potentially due to the increased

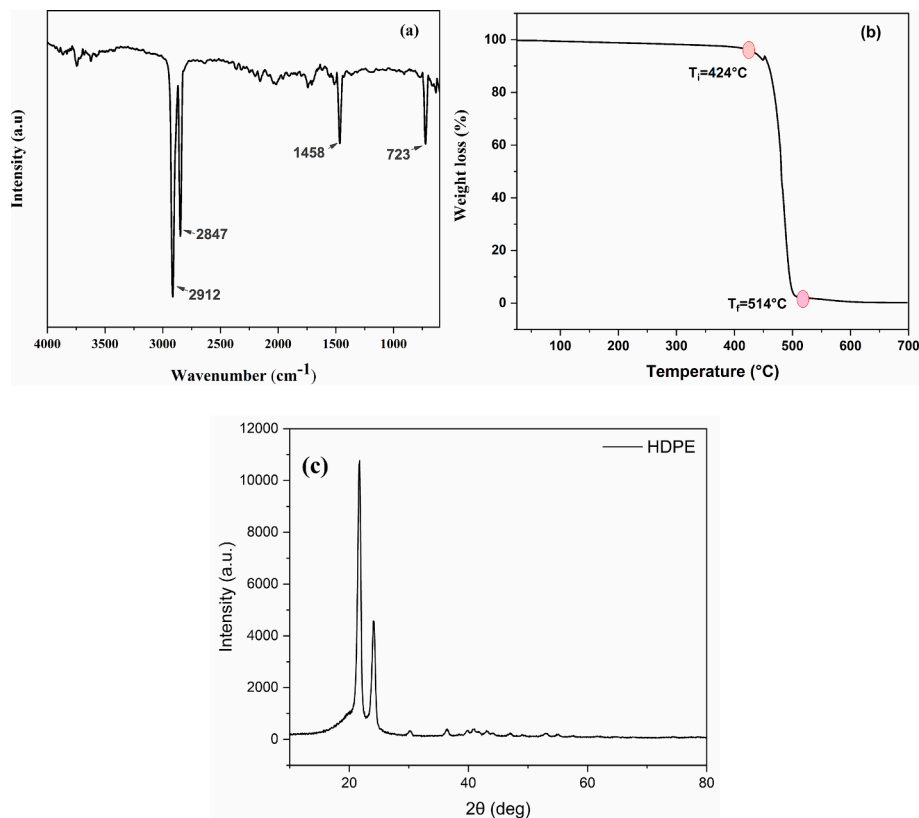


Fig. 7. FTIR spectra (a), TGA curve (b), and XRD (c) of Plastic feedstock spectra.

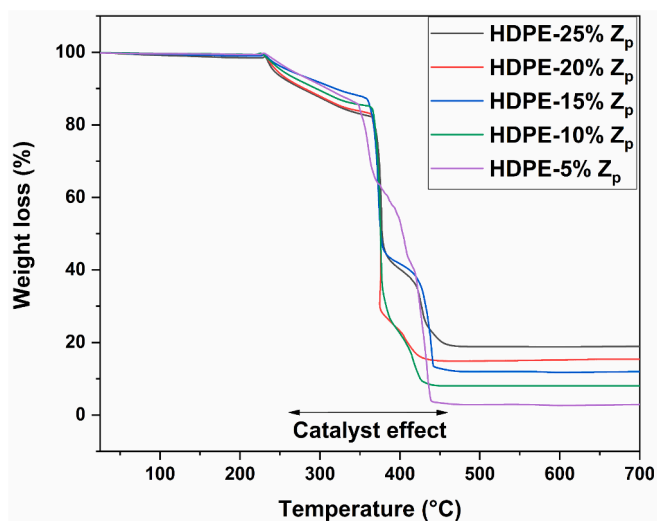


Fig. 8. TGA curve of HDPE in the presence of various amounts of synthesized zeolite.

HDPE adsorption facilitated by higher catalyst concentrations, leading to continued fragmentation with increasing temperature. The addition of 10 % synthesized catalyst enabled the complete transformation of HDPE at 419 °C, compared to 514 °C without the catalyst, representing a 20 % decrease. This result highlights the potential of P-Zeolite as a competitive option in the field of plastic waste pyrolysis, showing more promise than those reported in the literature for ZSM-5, which achieved a decrease of only 50 to 60 °C [51–52].

In order to better understand the effect of catalyst on the catalytic pyrolysis, Table 5, provides a comprehensive comparison of various zeolite types based on key parameters such as S_{BET} (specific surface area), pore volume, Si/Al ratio, acidity (mmol NH₃/g), and the saved temperature during catalytic processes. It clearly indicates a relationship between the Si/Al ratio and acidity for all catalysts. Zeolites with lower Si/Al ratios, such as HY (5.4 %), exhibit higher acidity values (0.99 mmol NH₃/g for HY), suggesting that a lower Si/Al ratio correlates with increased acidity. Conversely, zeolites with higher Si/Al ratios, like ZSM-5 (50 %) and MCM-41 (pure silicon), have lower acidity values (0.176 mmol NH₃/g and 0.12 mmol NH₃/g, respectively), which suggests that the tested zeolite likely has higher acidity than the other zeolites. The results of the table show that despite MCM-41 and HY zeolites having the largest specific surface areas, their contributions to

Table 4

Degradation temperature of HDPE in the absence and the presence of various catalyst amount.

Peak degradation Sample Code	1st peak degradation			2nd peak degradation			3rd peak degradation		
	T _{onset} (°C)	T _{Fin} (°C)	Wt. loss (%)	T _{onset} (°C)	T _{Fin} (°C)	Wt loss (%)	T _{onset} (°C)	T _{Fin} (°C)	Wt loss (%)
HDPE	424	514	99	***	***	***	***	***	***
HDPE-5 %Z	234	350	8	342	401	44	414	440	45
HDPE-10 %Z	234	358	9	362	382	69	391	419	13
HDPE-15 %Z	234	361	15	363	393	51	400	448	27
HDPE-20 %Z	234	360	18	364	378	58	382	438	13
HDPE-25 %Z	234	361	14	362	386	47	394	458	19

Table 5
Intrinsic Properties of Reported Zeolites and Their Influence on Pyrolysis Temperature.

Zeolite type	S_{BET} ($\text{m}^2 \cdot \text{g}^{-1}$)	Pore volume ($\text{cm}^3 \cdot \text{g}^{-1}$)	Si/Al ratio(%)	Acidity $\text{mmol NH}_3 \cdot \text{g}^{-1}$	Saved temperature($^{\circ}\text{C}$)	References
ZSM-5	412.0	0.397	50	0.176	60	[53]
MCM-41	1105	1.08	Pure silicon	0.12	26	[54]
HY	625	0.33	5.4	0.99	226	[54]
H β	489	0.41	25	0.50	52	[54]
Tested zeolite	134	0.81	2.08	***	100	This study

temperature reduction are significantly different. HY zeolite has proven capable of reducing the temperature by 226 $^{\circ}\text{C}$, whereas MCM-41 only reduces it by 60 $^{\circ}\text{C}$. This is largely due to the difference in acidity between the two materials. A similar observation can be made when comparing ZSM-5 and H β , which have nearly the same specific surface areas but very different Si/Al ratios, leading to significantly different acidities. H β exhibits the highest acidity and, consequently, the greatest temperature reduction. These results demonstrate that the Si/Al ratio, which is directly linked to the local acidity of the zeolites, is the most influential parameter in temperature reduction. Additionally, these findings corroborate the results obtained in this study.

The Criado method was employed to investigate the kinetics of HDPE degradation, providing valuable insights into its thermal behavior. This widely used approach facilitates the determination of crucial parameters like activation energy and reaction order. Through this analysis, we aim to uncover the degradation mechanism of HDPE and factors impacting its thermal stability. This study explores the variations in HDPE degradation kinetics under different conditions, offering vital data to optimize processing parameters and improve the polymer's performance across diverse applications. Initially, Criado's and Coats–Redfern methods were employed to discern the most suitable reaction mechanisms, utilizing master plots illustrated in Fig. 9. Subsequently, based on these findings, the most appropriate kinetic mechanism was determined for each test. Notably, plots of kinetic mechanisms D1, R1, P2, P3, and P4 were omitted due to their comparatively weak performance compared to other models. The most effective reaction model, identified with the highest mean coefficient, is D2 with $R^2 = 0.93$. Previous studies suggest that the optimal reaction models for HDPE are the contracting cylinder models (R2) and (D2). Specifically, the R2 model is predominantly observed at higher heating rates exceeding 10 $^{\circ}\text{C}/\text{min}$ [35].

After determining the model corresponding to the HDPE degradation mechanism, we proceeded to solve the thermal degradation equation of

HDPE using the KAS method. Thermogravimetric analyses were conducted in both the presence and absence of a 10 % zeolite catalyst at different heating rates (10 $^{\circ}\text{C}/\text{min}$, 15 $^{\circ}\text{C}/\text{min}$, and 20 $^{\circ}\text{C}/\text{min}$). The analysis of the TGA curves (Fig. 10 a, b) confirms that the use of the zeolite catalyst modifies the HDPE degradation mechanism, and increasing the heating rate accelerates the thermal degradation phenomenon. The KAS model was used to obtain the kinetic parameters (E_a and A_0) by plotting $\ln[\beta/T^2]$ against $1/T$ for the different heating rates (Fig. 10 c, d), where the slope of the lines allows deduction of E_a .

Table 6 presents the values of E_a for each heating rate. The values of E_a range from 288 kJ/mol to 163 kJ/mol for HDPE without and with 10 % synthesized zeolite, respectively. The addition of synthesized zeolite during the pyrolysis process contributes to reducing the activation energy from 288 kJ/mol to achieve 163 kJ/mol, representing a reduction of 43.5 %, this decrease will contribute to lowering the required energy for HDPE pyrolysis. Similar finding has been reported by various researchers.

3.4. Effect of catalyst on the decomposition enthalpy of HDPE

The impact of a catalyst on the decomposition enthalpy of HDPE was investigated through DSC experimental analysis conducted in the presence and absence of the catalyst (Fig. 11). The DSC curves indicate two peaks related to the melting and decomposition steps. The melting temperature was recorded at 143 $^{\circ}\text{C}$ for both samples, with melting enthalpies of 184.73 J/g for HDPE without the catalyst and 200.12 J/g with the catalyst. Furthermore, as shown in Table 7, the enthalpy of degradation increased by 150 J/g in the presence of the catalyst compared to thermal degradation processes. It is hypothesized that the presence of the catalyst could potentially alter the rate or mechanism of HDPE decomposition, thereby influencing the reaction's enthalpy. This exploration aims to deepen our understanding of how catalysts affect the thermal stability and degradation characteristics of HDPE, providing valuable information for optimizing catalytic processes and improving the efficiency of plastic waste recycling endeavors. As illustrated in Fig. 11, the synthesized zeolite catalyst demonstrated a noticeable shift in T_{max} , accompanied by a curve depicting endothermic behavior. The synthesized zeolite catalyst enhances the degradation enthalpy of HDPE plastic by making bond cleavage more efficient, leading to a more complete breakdown of the polymer chains and resulting in a higher total energy release during decomposition. Moreover, the presence of the catalyst promotes the formation of more stable and lower-energy products, increasing the energy difference between the reactants and the products, thereby raising the enthalpy change.

4. Conclusion

The synthesis of the zeolite catalyst was successfully achieved, as confirmed by various structural and morphological characterization techniques. The catalyst exhibited enhanced thermal stability up to 800 $^{\circ}\text{C}$ and, with a catalyst content of 10 % wt, effectively lowered the plastic degradation temperature. Moreover, the presence of the catalyst induced a significant change in the degradation mechanism, leading to more pronounced bond cleavage compared to the absence of the catalyst. Theoretical simulations using numerical methods, such as the

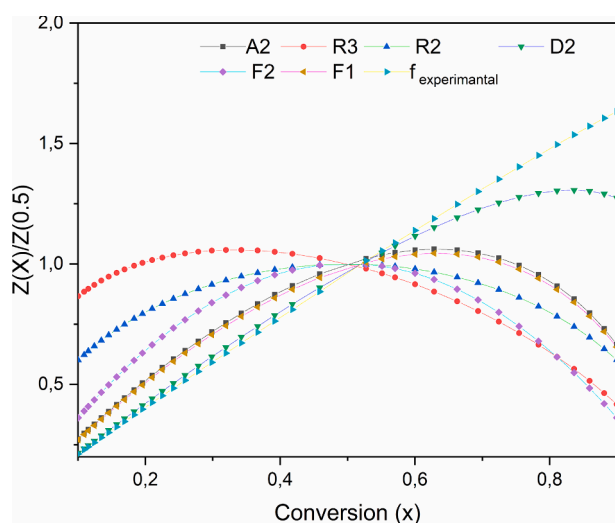


Fig. 9. Theoretical diagrams for different reaction models compared with experimental data for HDPE pyrolysis at β 10 $^{\circ}\text{C}/\text{min}$.

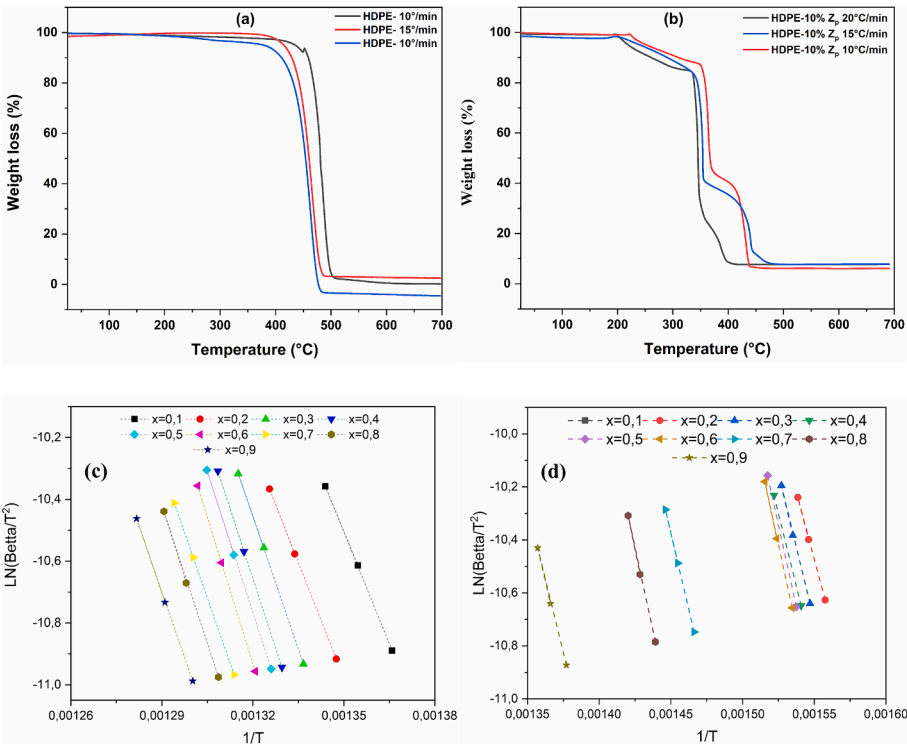


Fig. 10. (a) Thermogravimetric analysis (TGA) of HDPE degradation at different heating rates without the catalyst. (b) TGA of HDPE degradation at different heating rates with 10% of the synthesized catalyst. (c) Linear regression lines of HDPE cracking at different conversions using the KAS method without the catalyst. (d) Linear regression lines of HDPE cracking at different conversions using the KAS method with 10% of the synthesized catalyst.

Table 6
Activation energies using the KAS method.

Conversion	Activation energy without catalyst (kJ/mol)	Activation energy using catalyst (kJ/mol)
0.1	278	146
0.2	287	151
0.3	269	155
0.4	294	160
0.5	275	166
0.6	269	173
0.7	293	172
0.8	277	168
0.9	286	174
Average	280.88	162.77

Criado model, facilitated the determination of the HDPE degradation equation. By applying the KAS method, we quantified the activation energy required for HDPE cracking with and without the 10 % wt

catalyst. The analysis revealed that the catalyst reduced both the HDPE degradation temperature and the activation energy required for degradation by over 40 %. Thermal analysis by DSC further corroborated these findings, showing that HDPE degradation in the presence of the catalyst produced an enthalpy of 2172 J/g, representing an increase of over 30 % compared to degradation without the catalyst. These results highlight the critical role of catalysts in modifying the thermal stability and degradation behavior of polymers, offering important insights for optimizing catalytic processes and advancing plastic waste recycling

Table 7
Melting and Decomposition Temperatures and Heats from DSC Measurements.

Material	Melting		Decomposition	
	T/°C	Q _m /J g ⁻¹	T/°C	Q _d /J g ⁻¹
HDPE	143	184.73	434	2025.39
HDPE@P-Zeolite	145	200.12	368	2172.19

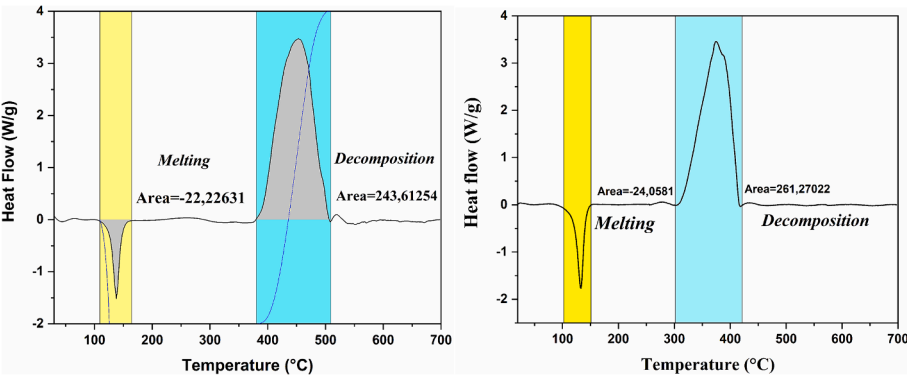


Fig. 11. DSC heating thermograms for HDPE and HDPE@P-Zeolite.

efforts. These findings pave the way for further investigation into practical scale-up considerations and the analysis of obtained products.

CRedit authorship contribution statement

Ilyass Belrhazi: Writing – original draft, Software, Investigation, Formal analysis, Data curation. **Said Sair:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration. **Hanane Ait Ousaleh:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Data curation, Conceptualization. **Youness Abdellaoui:** Writing – review & editing, Visualization, Validation, Resources, Investigation. **Mohamed Zahouily:** Writing – review & editing, Validation, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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